Tutorial 1

CSN-351/AID-523 Database Management Systems

1. List the details that are described and the details that are hidden by the three levels of abstraction with the help of an example. [Topic: Levels of abstraction]

2. Give an example of simple database schema and provide examples for each of the following terms [Topic: Relational model]
    a. Relation(table)
    b. Schema
    c. Tuple
    d. Attribute
    e. Degree of relation

3. Choose the correct statement(s) regarding superkeys. (One or more than one options may be correct) [Topic: Keys]
   a) A superkey is an attribute or a group of multiple attributes that can uniquely identify a tuple
   b) A superkey is a tuple or a set of multiple tuples that can uniquely identify an attribute
   c) Every candidate is a superkey key
   d) A superkey is an attribute or a set of attributes that distinguish the relation from other relations

4. Mention any three differences between primary key and candidate key. [Topic: Keys]

5. Let R = (A, B, C, D, E, F) be a relation scheme with the following dependencies-

$$\{C\} \rightarrow \{F\}$$
$$\{E\} \rightarrow \{A\}$$
$$\{E,C\} \rightarrow \{D\}$$
$$\{A\} \rightarrow \{B\}$$

Which of the following is a key for R?

a. {C,D} b. {E,C} c. {A,E} d.{A,C}

[Note: X → Y indicates that attribute (or set of attributes) X uniquely identifies attribute (or set of attributes) Y]

6. Consider the relation scheme R(E, F, G, H, I, J, K, L, M, N) and the set of functional dependencies-

$$\{ E, F \} \rightarrow \{ G \}$$
$$\{ F \} \rightarrow \{ I, J \}$$
$$\{ E, H \} \rightarrow \{ K, L \}$$
$$\{ K \} \rightarrow \{ M \}$$
$$\{ L \} \rightarrow \{ N \}$$

Which of the following is the key for R?

a. { E, F }
b. { E, F, H }
c. { E, F, H, K, L }
d. { E }

Also, determine the total number of candidate keys and super keys.